
Installing Dell OpenManage Essentials

This technical white paper describes how to install OpenManage Essentials.

OME Engineering Team



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December 2014 | Rev 1.4

Contents

Executive Summary 5

What is OpenManage Essentials? 5

Minimum Installation Requirements..... 5

Installation Procedures 7

 Installation Flowchart..... 7

 Installing Dell OpenManage Essentials..... 9

 Critical Prerequisites..... 10

 Automatically Installing All Critical Prerequisites 11

 SQL Server Instance Selection..... 12

 Typical Installation 13

 Custom Installation 13

 Remote Database Installation 14

 Installing OpenManage Essentials on a Domain Controller 16

 Configuring Managed Service Account 16

 Configuring SQL Server on a Domain Controller 17

 Configuring OME Groups 18

OpenManage Essentials Supporting Utilities 20

 Troubleshooting Tool..... 20

 Dell OpenManage Essentials MIBImport Utility 20

Upgrade Workflow Flowchart 21

 Data Migration during Upgrade..... 22

Uninstalling 24

Repair (Program Maintenance)..... 25

Learn more 26

Conclusion 26

Tables

Table 1. Minimum recommended hardware 5

Table 2. Minimum software and network requirements..... 6

Table 3. Minimum login roles for Microsoft SQL Server 6

Table 4. Login roles..... 7

Table 5. Local Database Instances Supported..... 12

Table 6. Default port settings..... 14

Table 7. Migration on Local database 22

Figures

Figure 1. Fresh installation..... 8

Figure 2. Dell OpenManage Install Available options dialog box..... 9

Figure 3. Dell OpenManage Essentials prerequisites..... 10

Figure 4. SQL Server Warning prerequisite 11

Figure 5. Port settings..... 13

Figure 6. Database server window from the InstallShield Wizard. 14

Figure 7. Browsing a remote database. 15

Figure 8. Managed Service Account..... 16

Figure 9. Configuring SQL Server 17

Figure 10. Providing credentials for the account 18

Figure 11. Adding Users to OME Groups 19

Figure 12. Upgrade workflow..... 21

Figure 13. Ready to Install the Program screen..... 23

Figure 14. Custom settings (only displayed during the upgrade process). 24

Figure 15. Uninstalling OpenManage Essentials. 25

Figure 16. Retain Database Option while Uninstalling 25

Figure 17. Program maintenance. 26

Executive Summary

This document describes the Dell OpenManage Essentials (OME) installation prerequisites, installation flow and steps, and information you can make use of during the OME installation or upgrade from previous versions.

This document includes the following information:

- Minimum hardware and software requirements
- Single-click prerequisites installation
- Remote database installation
- Program maintenance

What is OpenManage Essentials?

Dell OpenManage Essentials is a lightweight, Web-based, one-to many systems management solution that provides a comprehensive view of Dell systems, devices, and components in an enterprise network. To know more refer: www.delltechcenter.com/ome

Minimum Installation Requirements

Installing OpenManage Essentials requires the minimum software, network, and system requirements detailed in the tables below. You must have system administrator privileges to perform the installation.

Table 1. Minimum recommended hardware

Minimum recommended Hardware	Large deployments	Large deployments	Medium deployments	Small deployments
Number of devices	Up to 4000	Up to 2000	Up to 500	Up to 100
Type of system	Physical machines / virtual Machines	Physical machines / virtual Machines	Physical machines / virtual Machines	Physical machines / virtual Machines
RAM	8 GB	8 GB	6 GB	4 GB
Processors	8 cores total	8 cores total	4 cores total	2 cores total
Database	SQL standard	SQL standard	SQL Express	SQL Express
Database location	Remote	Remote	Local	Local
Hard drive	20 GB	10 GB	6 GB	6 GB

Table 2. Minimum software and network requirements

Particulars	Minimum requirements
Operating systems	Microsoft Windows Server 2008 SP2 Standard and Enterprise Editions (x64) Microsoft Windows Server 2008 R2 SP1 Standard and Enterprise Editions Microsoft Windows Server 2012 Standard and Datacenter Editions Microsoft Windows Server 2012 R2 Standard and Datacenter Editions NOTE: OpenManage Essentials version 2.0.1 is supported only on x64 operating systems.
Network	1 Gbps or higher
Web browser	Microsoft Internet Explorer 9, 10 and 11 Mozilla Firefox 22 and 23 Google Chrome 30 or 31
Database	Standard\Enterprise\Datacenter Editions: Microsoft SQL Server 2008 and above Express Editions: SQLEXPRESSOME instance with Microsoft SQL Server 2008 and above NOTE: OpenManage Essentials installation is supported only on a case-insensitive instance of SQL Server.
User interface	Microsoft Silverlight version 5.1. 30214.0
Microsoft .NET Framework	4.5
Microsoft Visual C++ Runtime	Runtime 11.0

Note: If Windows is not having the above mentioned Service Packs installed, installation of Essentials in any kind including Typical, Custom and upgrade from previous version will be blocked. You should adhere to the above mentioned Service Packs of Windows for installing/upgrading Essentials.

Table 3. Minimum login roles for Microsoft SQL Server

Use cases	Minimum login roles for SQL Server
Fresh OME installation: select the <i>Typical installation</i> option.	<i>sysadmin</i> access on the installed instance
Fresh OME installation: select the <i>Custom installation</i> option. An <i>empty</i> OpenManage Essentials database is already present (locally or remotely) on the mentioned database server.	<i>db_owner</i> access on the OpenManage Essentials database.
Fresh OME installation: select the <i>Custom installation</i> option, and no OME database found.	<i>dbcreator</i> access on the server.
Upgrading OME from v1.3 or v2.0 to 2.0.1 and an OpenManage Essentials database is present (locally or remotely).	<i>db_owner</i> access on the OpenManage Essentials database.

Note: If you select the *Custom* install option and do not enter any credentials (irrespective of whether the Windows or SQL Authentication method is selected), then the installation is considered as a “*Typical*” installation and “*sysadmin*” rights will be checked for the mentioned database server and database instance.

Table 4. Login roles

Role	Why does OME require this role during installation?
Sysadmin	During a Typical install, OME requires this role to create a new SQL Server Login <i>OMEService</i> . This login is granted with <i>dbcreator</i> permission.
dbcreator	During a Custom install, OME requires this role to create the OMEssentials database on the mentioned database server.
db_owner	OME requires this role when it detects an existing OMEssentials database as present on the system. This role helps OME to create/update the OME database objects on this existing database.

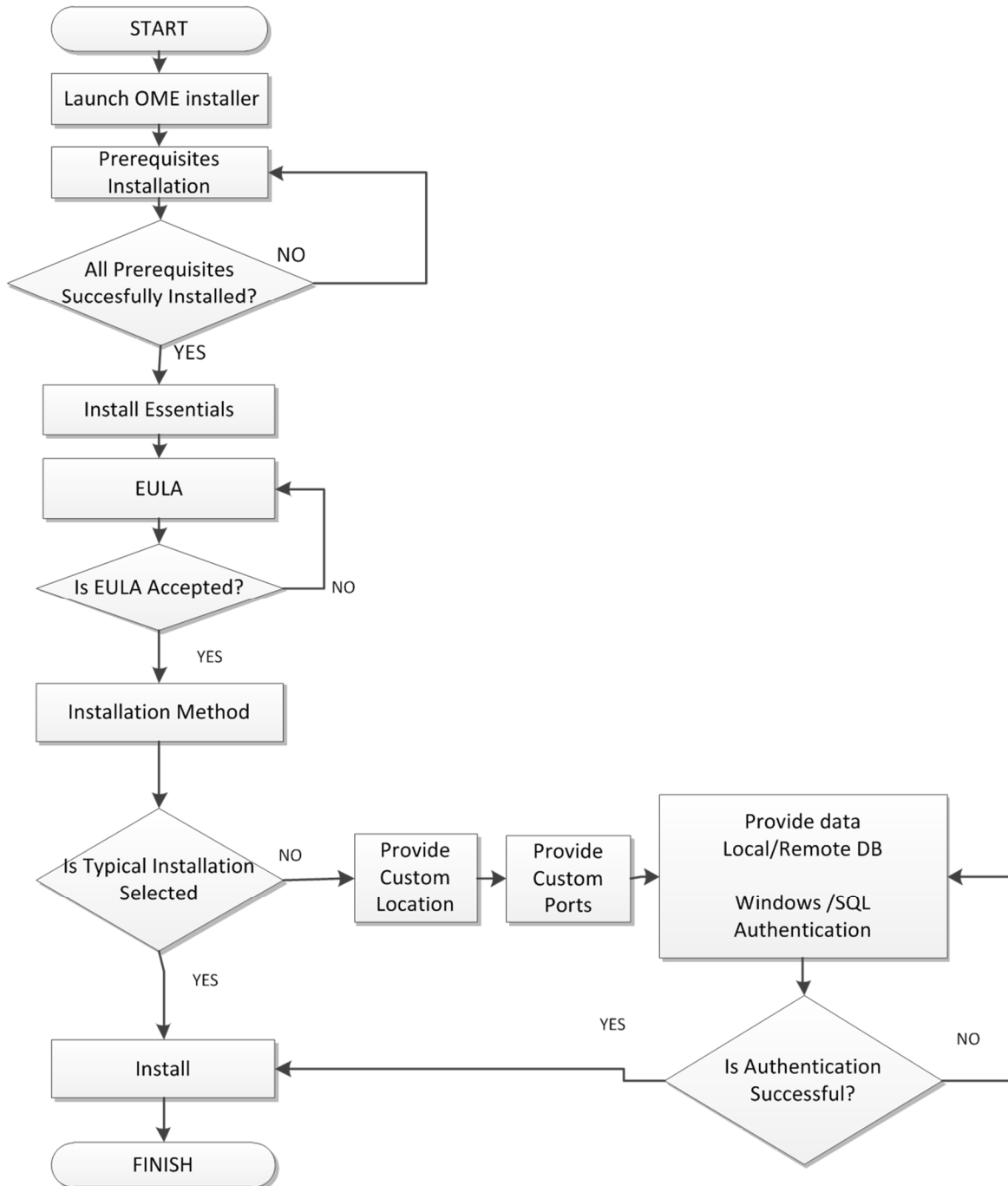
Installation Procedures

Use the following install procedures for installing OpenManage Essentials.

Installation Flowchart

The Figure 1 shows the brief OME installation steps with respect to a fresh installation.

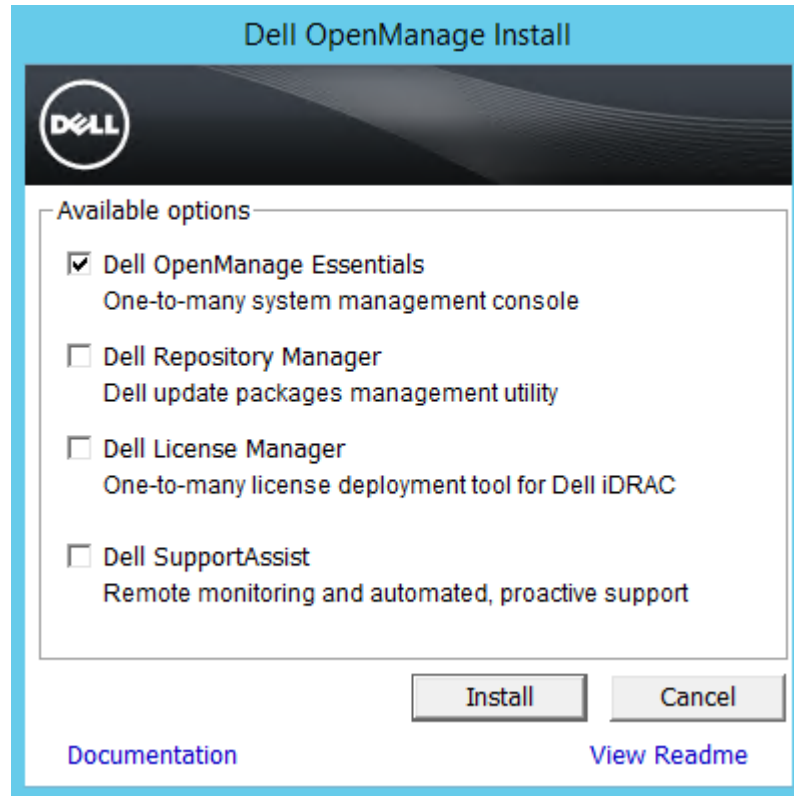
Figure 1. Fresh installation



Installing Dell OpenManage Essentials

1. Download the OpenManage Essentials package from <https://delltechcenter.com/ome>, and extract the files to a local drive.
2. The self-extracting zip automatically launches the Autorun.exe window or you can manually launch the Autorun.exe.

Figure 2. Dell OpenManage Install Available options dialog box.

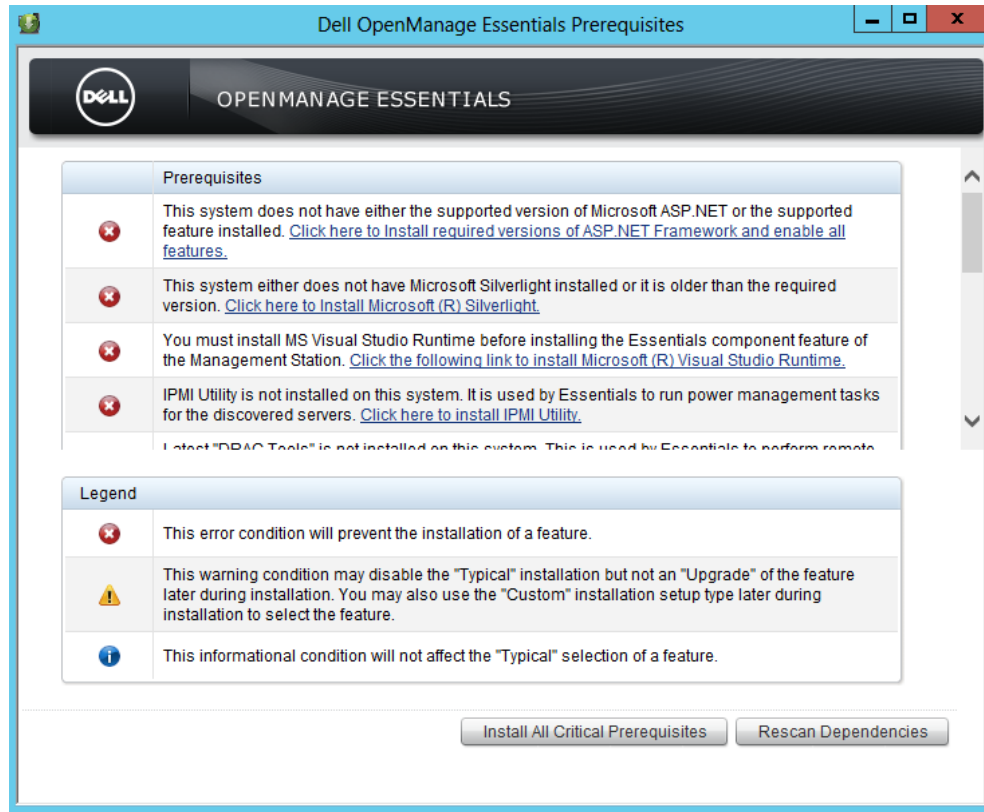


3. Select **Dell OpenManage Essentials**, and then click **Install**. This launches the Dell OpenManage Essentials Prerequisites screen.

Note:

- When you select OME, the *Dell Troubleshooting Tool* and the *Dell OME MIBImport Utility* are installed along with it.
- If you directly launch the Setup.exe in the \windows\ManagementStation\ folder, then the *Dell Troubleshooting Tool* and the *Dell OME MIBImport Utility* are not installed along with OME.

Figure 3. Dell OpenManage Essentials prerequisites.



Critical Prerequisites

If a component that is required for OpenManage Essentials installation is missing or if the version is outdated, it displays as a critical prerequisite on the **Dell OpenManage Essentials Prerequisites** screen (Figure 3). It means you cannot install OME unless all these requirements are met on the system.

The critical prerequisites are:

- IIS 7.5/8
- Microsoft .NET 3.5 SP1
- ASP.NET 4.5
- Microsoft Silverlight 5.1
- Microsoft Visual Studio® Runtime 11.0
- BMC Utilities 4.6.0
- DRAC Tools 8.0.1
- PowerShell 2.0

Note: OpenManage Essentials installation is not dependent on the Windows SNMP service. The OpenManage Essentials installer will not prevent the installation if you have the SNMP service disabled or stopped. If the SNMP Trap service is configured in manual mode and disabled, the Prerequisites window displays a warning prerequisite. If you ignore this warning and proceed with the installation, OpenManage

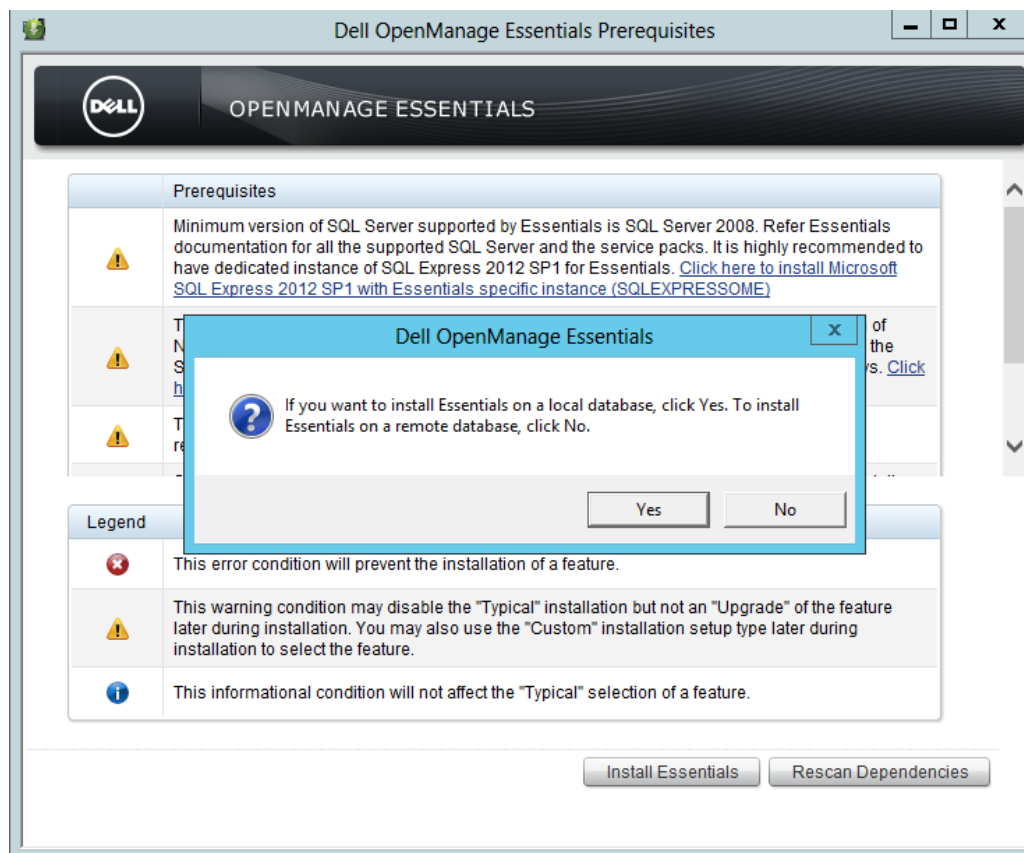
Installing Dell OpenManage Essentials

Essentials will not be able to receive SNMP traps/alerts. However, if you want to enable the SNMP traps/alerts reception functionality at later time, follow these steps:

1. Manually enable and start the Windows SNMP Trap service from Service Control Manager.
2. In OpenManage Essentials, navigate to “Preferences” → “Console Settings”.
3. Click “Restart Services”.

Note: If the SQL server is not present on the system where OME is being installed a warning message is displayed, with a link to install SQLEXPRESS locally. If you ignore this warning message, then while launching the Essentials installation you will be asked whether you want to install on local or remote database. If you want to go with installing Essentials on a local database, then select “Yes”. On doing so, SQLEXPRESS will be installed first with the SQLEXPRESSOME instance name. If you wish to install Essentials on a remote database, please select “No”. On doing so, the installer will follow the Custom installation path where you have to select a remote database for installing Essentials. Please refer the following figure:

Figure 4. SQL Server Warning prerequisite



Automatically Installing All Critical Prerequisites

To automatically install all critical prerequisites, click *Install All Critical Prerequisites* button on the *Dell OpenManage Essentials Prerequisites* screen. Depending on the system configuration, a reboot may be

Installing Dell OpenManage Essentials

required; however, the prerequisites installation resumes automatically after the system restarts. To avoid issues with automatically resuming installation, make sure that *User Access control* is turned off.

Note: The installation may take up to 35 minutes to complete.

SQL Server requirements include:

- **Local Database:**
To set up OpenManage Essentials on a local database, SQL Server 2008/2008 R2/2012 Standard/Enterprise/Datacenter edition or SQL Server 2008/2008 R2/2012 Express edition with SQLEXPRESSOME instance name must be installed (even though it displays only as a warning prerequisite).
- **Remote Server:**
If you have a database located on a remote server, SQL Server is *not* required on the local system.
- **Version:**
If the local system has SQL Server 2005 installed, it is recommended to install SQL Server 2008 or SQL Server 2012. OpenManage Essentials is not supported on SQL Server 2005.
- **Collation:**
OpenManage Essentials is supported only on a case-insensitive instance of SQL Server.

Note: SQL Server 2012 RC/Evaluation versions are not supported.

SQL Server Instance Selection

Different instances of SQL Server will be selected by Essentials installation in different scenarios. Automatic instance selection is different when Express or Standard/Enterprise/Datacenter editions of SQL Server are installed on the local system. If Essentials is targeted to be configured on a remote database, you can select any instance of SQL Server 2008/2008 R2/2012 Standard/Enterprise/Datacenter/Express editions. Please refer the following table to know more about the instance selection logic of the Essentials installer during “Typical” install.

Table 5. Local Database Instances Supported

SQL Version	SQL Edition	Instance Name	Prerequisite Status	Prerequisite Message	Required Action
None	None		Warning	No SQL found	Click to install
2005	Standard/Enterprise	Default	Warning	No supported SQL found	Click to install
		Custom Name	Warning	No supported SQL found	Click to install
2005	Express	Default	Warning	No supported SQL found	Click to install
		Custom Name	Warning	No supported SQL found	Click to install
2008/R2	Standard/Enterprise/Datacenter	Default/ Custom Name	None	None	None
		SQLEXPRESSOME	None	None	None
2008/R2	Express	Default/ Custom Name	Warning	No supported SQL found	Click to install
		SQLEXPRESSOME	None	None	None
2012	Standard/Enterprise	Default/ Custom Name	None	None	None
		SQLEXPRESSOME	None	None	None
2012	Express	Default/ Custom Name	Warning	No supported SQL found	Click to install

		SQLEXPRESSOME	None	None	None
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Typical Installation

A Typical installation installs OME with the default settings. Select *Typical* on the Setup Type screen, and then click Install on the Ready to Install the Program screen.

In a Typical installation:

- SQL Server Authentication is selected as the default authentication method.
- The default SQL instance installed with SQL Server Standard/Enterprise/Datacenter edition is selected as the database server.
- The SQLEXPRESSOME instance will be selected if only Express edition of SQL Server is installed on the system. No other instance names are supported with Express editions of SQL Server (version 2008 or higher).
- If there is no default SQL instance and multiple named instances of SQL installed with the SQL Server Standard/Enterprise/Datacenter edition exist, the first-installed named instance will be selected as the database server.

Custom Installation

Custom installation is recommended for remote database installation, or if you want to install OME with Windows Authentication, or if you want to use a non-default location and port settings.

Figure 5. Port settings

Dell OpenManage Essentials - Install Wizard

Dell OpenManage Essentials Custom Settings

Enter Custom Settings for Dell OpenManage Essentials.

Network Monitoring Service port number: 2606

Task Manager Service port number: 1279

Console Launch port: 2607

Package Server port: 1278

InstallShield

< Back Next > Cancel

Table 6. Default port settings

Port number	Protocol	Port type	Maximum encryption level	Direction	Usage
1278	HTTP	TCP	None	In/Out	Web GUI
1279	Proprietary	TCP	None	In/Out	Scheduling tasks
2606	Proprietary	TCP	None	In/Out	Network
2607	HTTPS	TCP	128-bit SSL	In/Out	Web GUI

To set up OpenManage Essentials on a local database with multiple SQL Server instances, select the server from the **Database Server** list (Figure 6).

Figure 6. Database server window from the InstallShield Wizard.

Remote Database Installation

Configure OpenManage Essentials to use a SQL server instance on a remote system. Before setting up the OpenManage Essentials database on the remote system, make sure the following prerequisites are met:

- The network communication between the OpenManage Essentials system and the remote database system is functioning.
- The SQL Server user has permission to backup, create, and configure databases.
- If you want to use SQL Authentication mode for remote database installation, then ensure “SQL Server and Windows Authentication Mode” is enabled on the remote SQL Server.
- TCP/IP is enabled on the remote system to avoid database connection issues.

Installing Dell OpenManage Essentials

To set up the OME database on the remote system, click *Browse* on the Database Server screen and select the remote system, or enter the hostname and database instance in the Database Server field (see Figure 7).

Perform the installation with one of the following authentication methods:

- **Windows Authentication**

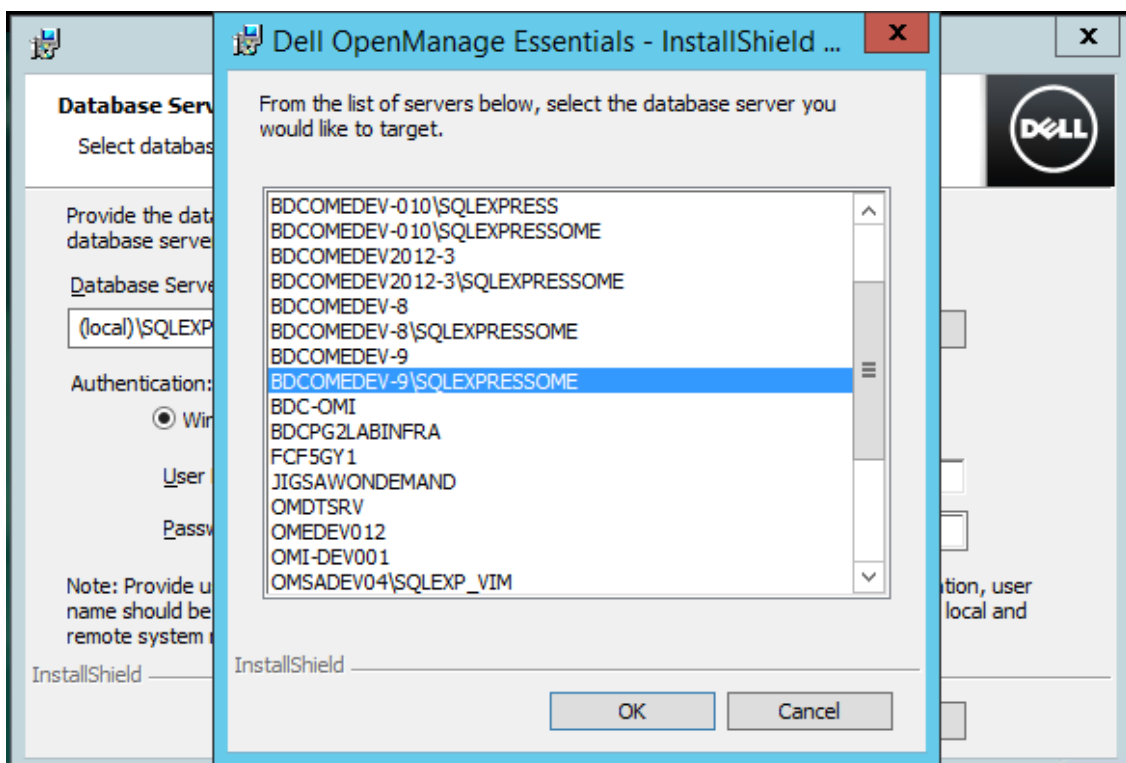
Windows login credentials are used to connect to the remote SQL Server. Use this authentication method if the Windows login credentials of the remote system and OME installation system are identical.

- **SQL Server Authentication**

Use this authentication method if the Windows login credentials of the remote database system and the OME installation system are different.

Note: After the OpenManage Essentials database is set up on the remote system, do not move the database. Otherwise, OpenManage Essentials stops working.

Figure 7. Browsing a remote database.



Installing OpenManage Essentials on a Domain Controller

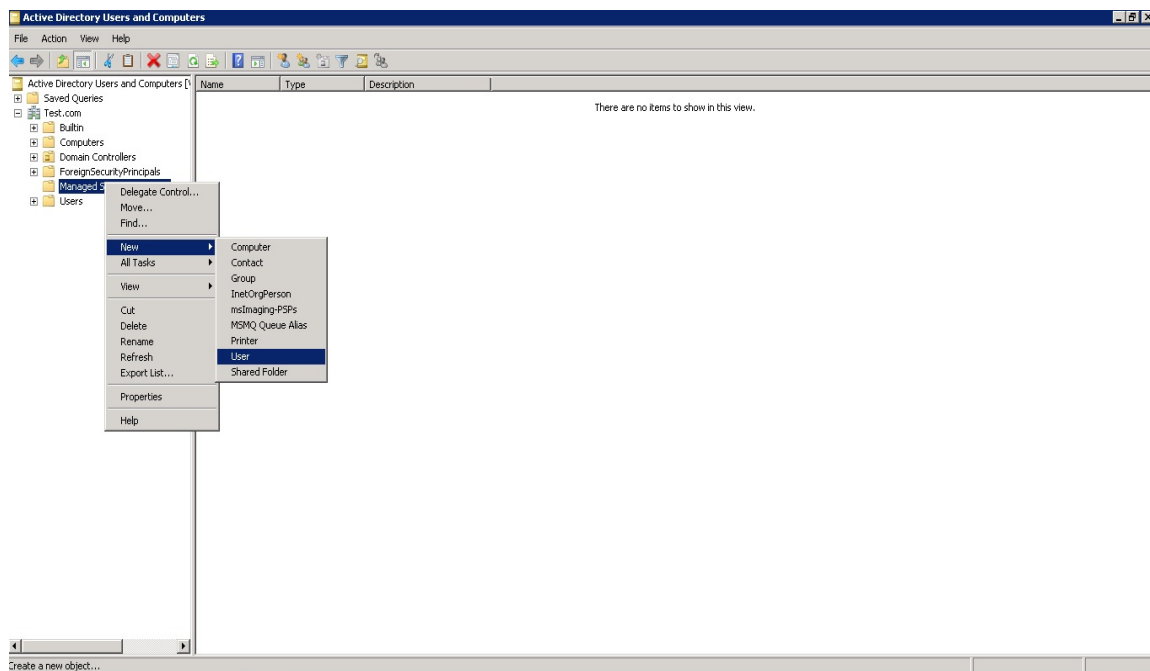
This section describes installing Dell OpenManage Essentials on a Domain Controller.

Configuring Managed Service Account

The following section provide procedures for configuring and using managed service accounts for installing OpenManage Essentials on a domain controller:

1. Go to "Start" → Administrative Tools.
2. Select "Active Directory Users and Computers".
3. On the left pane, right-click "Managed Service Account", and select "New" → "User" as shown in Figure 8.
4. Provide the First Name and user login name in the wizard.
5. Provide the password for the created user, confirm the password, and click Finish.
6. Add the user to the "Domain Admin" and "Remote Desktop User" groups.
7. Login to the system using same Service Account credentials.

Figure 8. Managed Service Account



Configuring SQL Server on a Domain Controller

It is recommended to install Microsoft SQL Server manually. If SQL Server is installed on the local machine, the SQL Server Service must be configured to run using a domain user account.

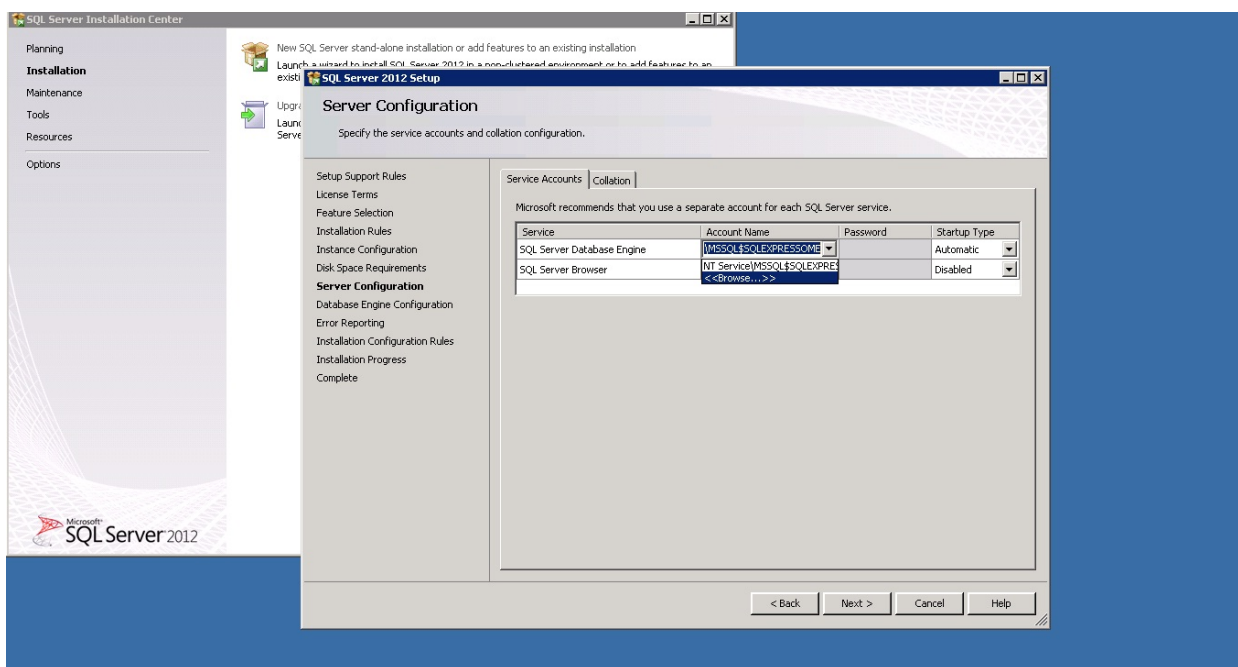
Note: The SQL Server Service will not start if you are using the default NETWORK SERVICE or LOCAL SYSTEM accounts.

Perform the following steps to set up the service accounts that will be used to install or run SQL Server.

Installing SQL on a domain controller:

1. Launch the SQL installer wizard.
2. After accepting the license terms, at the Instance Configuration wizard, provide “SQLEXPRESSOME” as the named instance.
3. Navigate till the “Server Configuration” wizard is displayed, as shown in Figure 9.

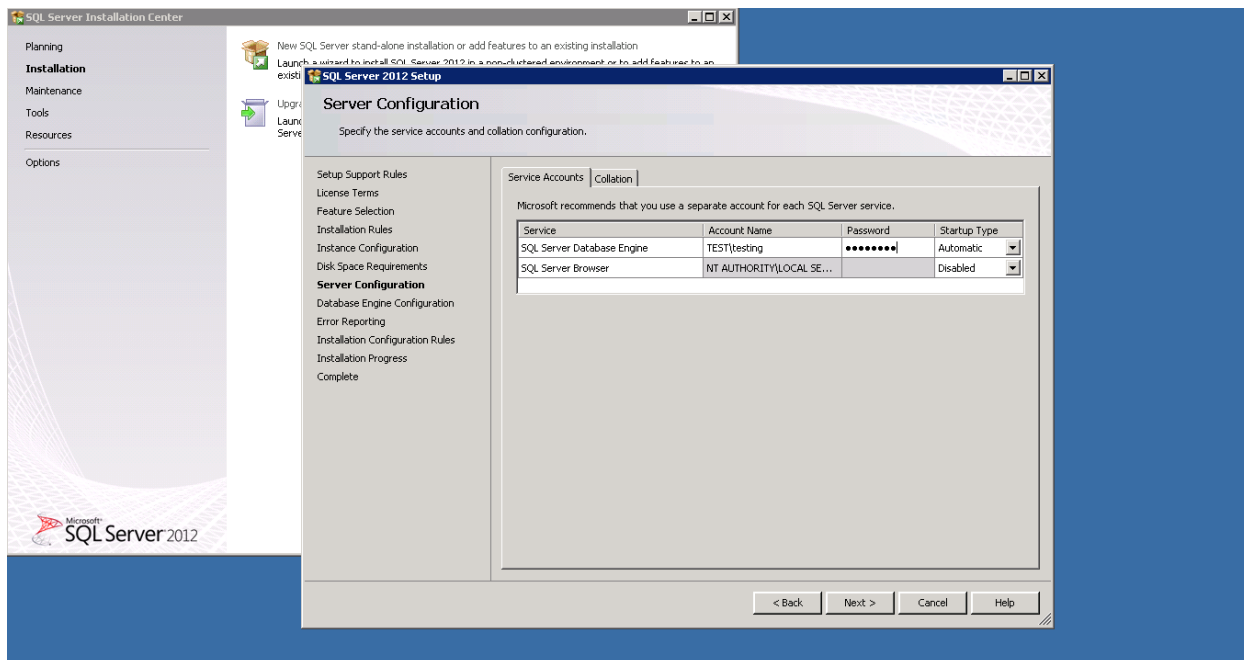
Figure 9. Configuring SQL Server



4. Click “Account Name” and select Browse.
5. Click “Advanced” and select “Find Now”.

6. Select the user in “Account Name” and provide the password, as shown in Figure 10.

Figure 10. Providing credentials for the account



7. At Database Engine Configuration, select “Mixed Mode”, provide the password and complete the installation.

Configuring OME Groups

After OpenManage Essentials is installed on a domain controller:

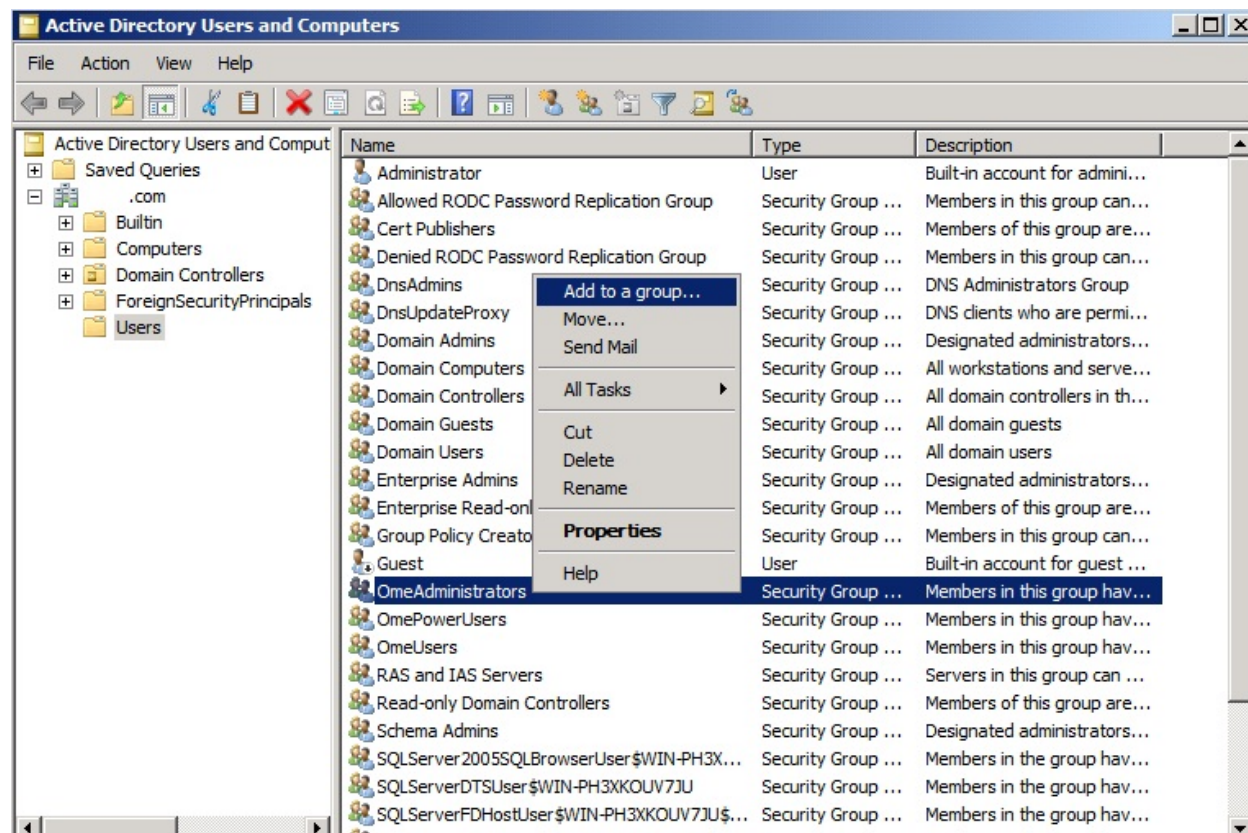
- By default, the Domain Admins group is added as a member of the OmeAdministrators, OmePowerUsers and OmeUsers groups.
- Local windows user groups are not included in the OpenManage Essentials roles. OMEAdministrators, OMEPowerUsers or OMEUsers privileges can be granted to user or user groups by adding them to the OpenManage Essentials windows groups. OMESiteAdministrators privileges can be granted by OMEAdministrators through the Device Group Permission portal.

To add users, perform the following steps:

1. Navigate to **Local Users and Groups** → **Groups**.
2. Right-click **OmeAdministrators** and select **Add to Group**.
3. In the **Properties** window, click **Add**.
4. In the **Select Users** window, provide the user name.
5. Click **Check Names** and click **OK**. The user name appears in the **Members** list in the **Properties** window.
6. Click **OK**.

Note: The users you are trying to add must also belong to the built-in local Administrators group. For details on adding a Windows user account to a group, see “Add a user account to a group” at <http://windows.microsoft.com/en-US/windows-vista/Add-a-user-account-to-a-group>

Figure 11. Adding Users to OME Groups



OpenManage Essentials Supporting Utilities

This section discusses the OpenManage Essentials supporting utility tools.

Troubleshooting Tool

A troubleshooting tool is automatically installed with OpenManage Essentials. This tool is useful for troubleshooting errors pertaining to supported device discovery and device connections.

If an older version of the Troubleshooting Tool exists on the system it gets upgraded as part of the OME installation. For more information, refer:

<http://en.community.dell.com/techcenter/systems-management/w/wiki/4172.dell-troubleshooting-tool.aspx>

Dell OpenManage Essentials MIBImport Utility

A MIB import utility is automatically installed with OpenManage Essentials. If you have non-Dell hardware, use this tool to define new alerts to OME. This lets you monitor a wider range of hardware and setup email and forwarding rules for these devices.

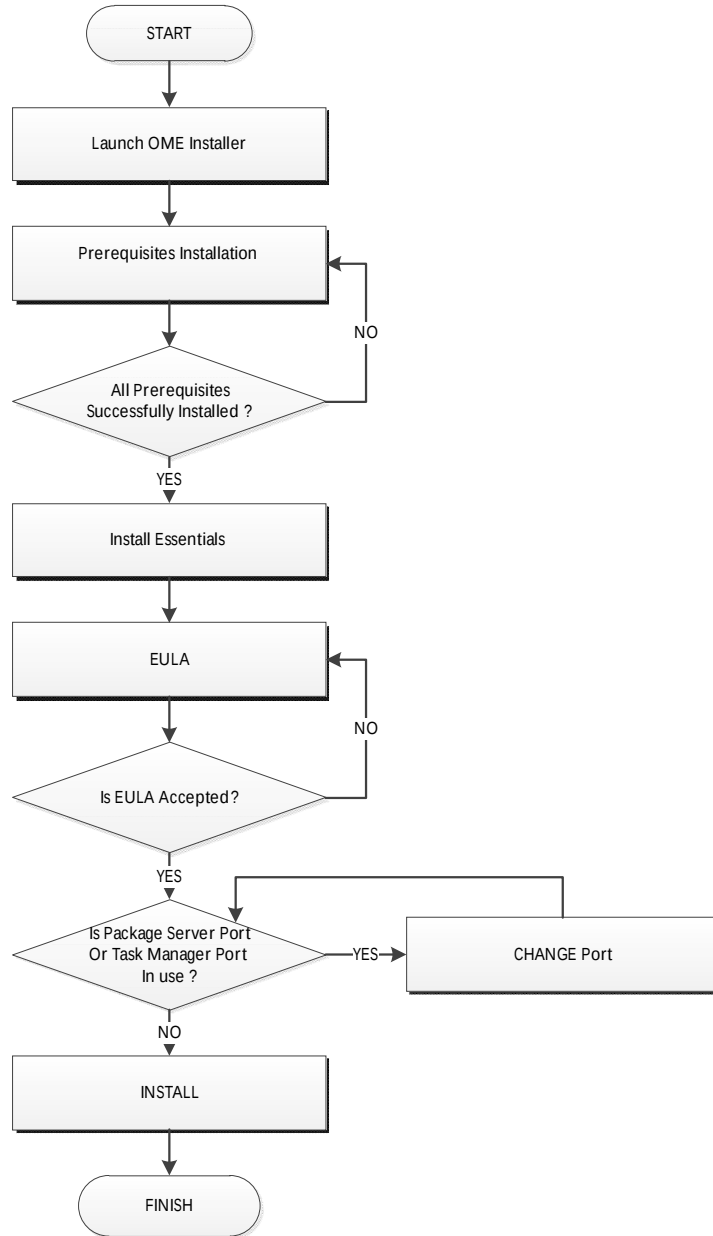
If an older version of MIB Import Utility exists on the system it gets upgraded as part of OME installation. For more information, go to:

<http://en.community.dell.com/techcenter/systems-management/w/wiki/3570.openmanage-essentials-mib-import-utility.aspx>

Upgrade Workflow Flowchart

The following figure displays the upgrade workflow.

Figure 12. Upgrade workflow



The OME upgrade retains the previous version data during upgrade process.

Note:

- You cannot upgrade directly from OME v1.0 (Beta Program) or v1.0.1 or v1.1 to OME v2.0.1. You must first upgrade to v1.3 or v2.0 and then to v2.0.1.
- OME cannot be upgraded to v2.0.1 if the previous version is installed on a **case-sensitive** instance of SQL Server targeted on a remote box or if it is the only instance running on the local box. In case the database is installed locally and a supported **case-insensitive** SQL Server instance is also found on the local box, the Essentials database will be migrated to a **case-insensitive** SQL Server instance during the upgrade.

During the upgrade, OME v2.0.1 uses the previously installed OME settings. These options (Custom Path, Ports, database server settings, and so on) are not displayed during an upgrade. The installation settings and configuration for v2.0.1 are displayed in the *Ready to install the Program* screen.

Data Migration during Upgrade

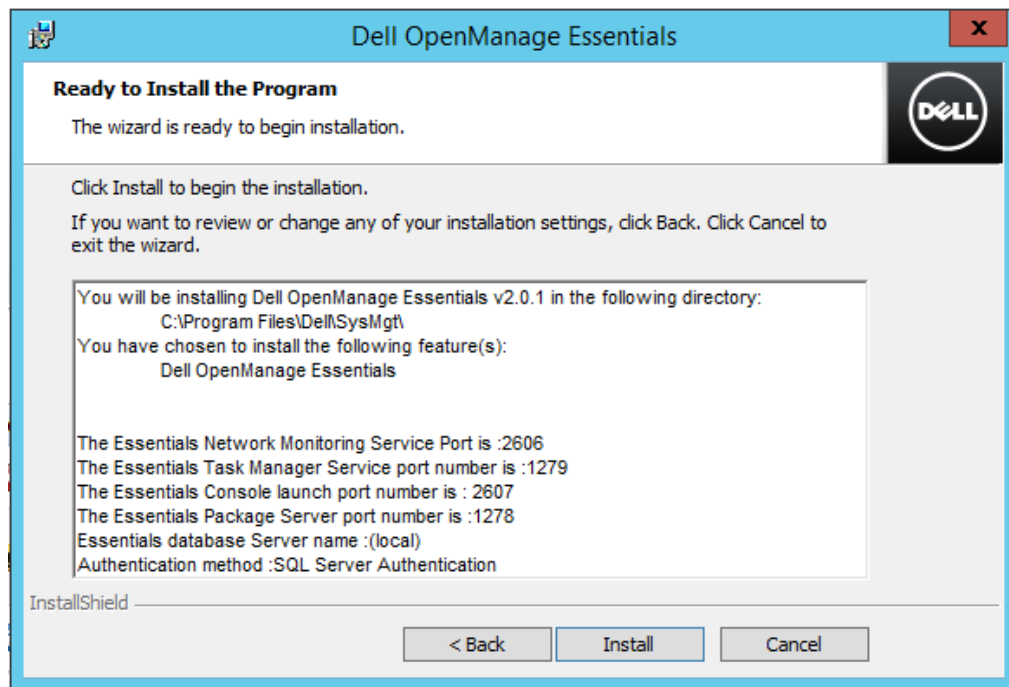
During upgrade, the Essentials database migration may happen if the SQL Server instance selected by v2.0.1 is different from the one used by the previous version of Essentials. Database migration will never happen during upgrade if the previous version of Essentials is installed on a remote database. Please refer Table 7 for probable cases of migration during upgrade to v2.0.1.

Example: Assuming you have installed v1.2 on local SQL Server Express edition with instance name SQLEXPRESS, then v2.0.1 will target the SQLEXPRESSOME instance installed with the SQL Server Express edition version 2012 (installed as part of the prerequisite) for installing the database during upgrade. Data from v1.2 will be copied to a new database “OMEssentials” created on the SQLEXPRESSOME instance, which will be used by v2.0.1. The old “OMEssentials” database on the SQLEXPRESS instance will be retained and will not be deleted.

Table 7. Migration on Local database

SQL Version	SQL Edition	Instance used for installing v1.0.1/v1.1	Prerequisite Status	Prerequisite Message	Data Migration Required?
2005	Standard/Enterprise	Default	Critical	No supported SQL found	Yes
		Custom Name	Critical	No supported SQL found	Yes
2005	Express	Default	Critical	No supported SQL found	Yes
		Custom Name	Critical	No supported SQL found	Yes
2008/R2	Standard/Enterprise/Datacenter	Default/Custom Name	None	None	No
		SQLEXPRESSOME	None	None	No
2008/R2	Express	Default/Custom Name	Critical	No supported SQL found	Yes
		SQLEXPRESSOME	None	None	No
2012	Standard/Enterprise	Default/Custom Name	None	None	No
		SQLEXPRESSOME	None	None	No
2012	Express	Default/Custom Name	Critical	No supported SQL found	Yes
		SQLEXPRESSOME	None	None	No

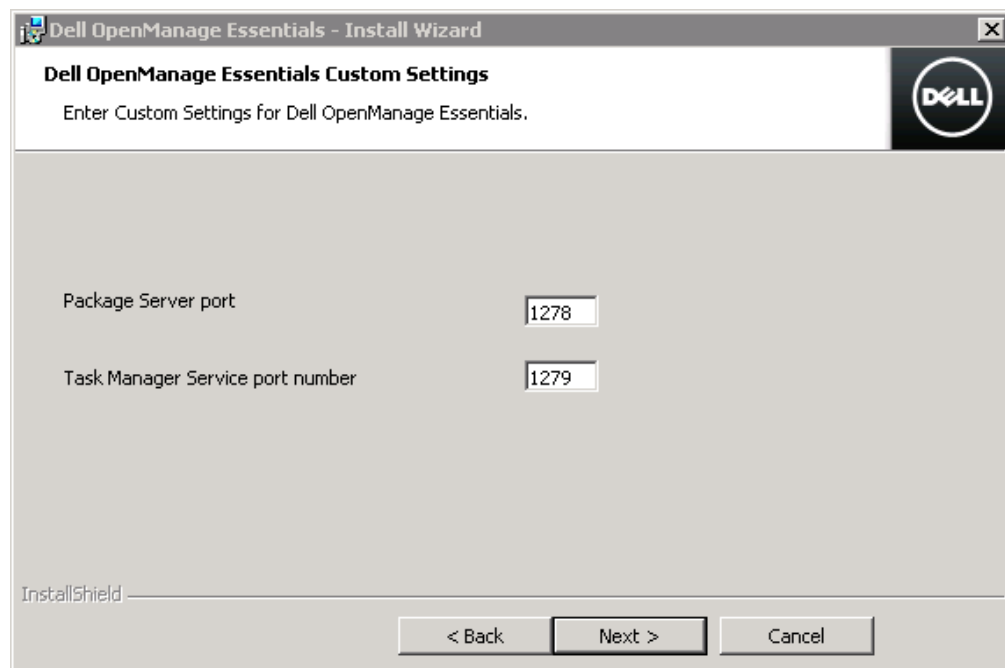
Figure 13. Ready to Install the Program screen.



You are prompted to change the ports (Figure 14) only if:

- The default Package Server Port 1278 is already in use/blocked.
- The default Task Manager Port 1279 is already in use/blocked.

Figure 14. Custom settings (only displayed during the upgrade process).

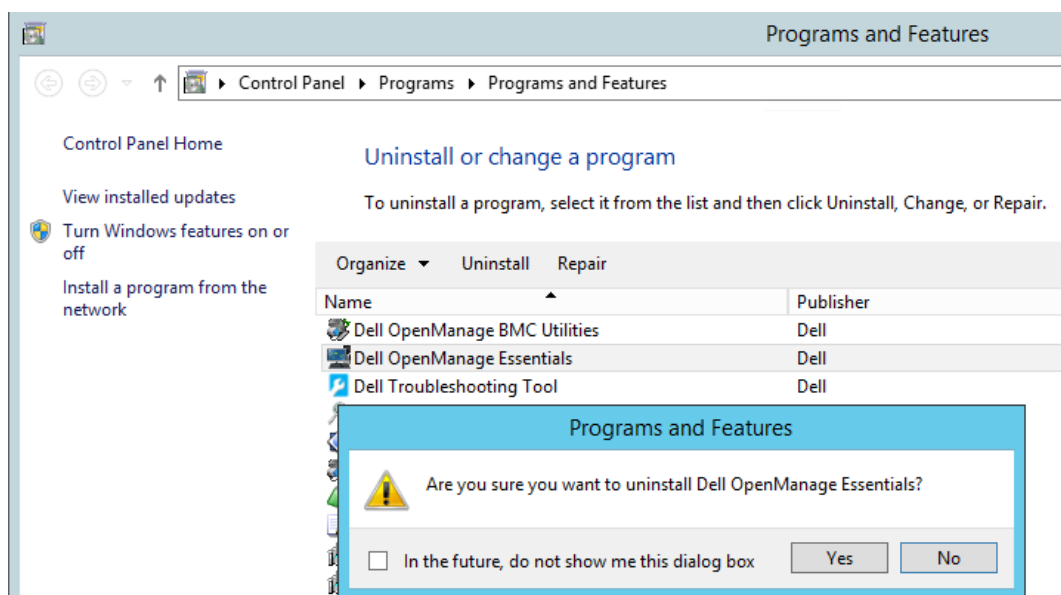


Uninstalling

Use this procedure to uninstall OpenManage Essentials:

1. Navigate to **Control Panel → Programs → Programs and Features** and select **Dell OpenManage Essentials**.
2. Click **Uninstall**.

Figure 15. Uninstalling OpenManage Essentials.

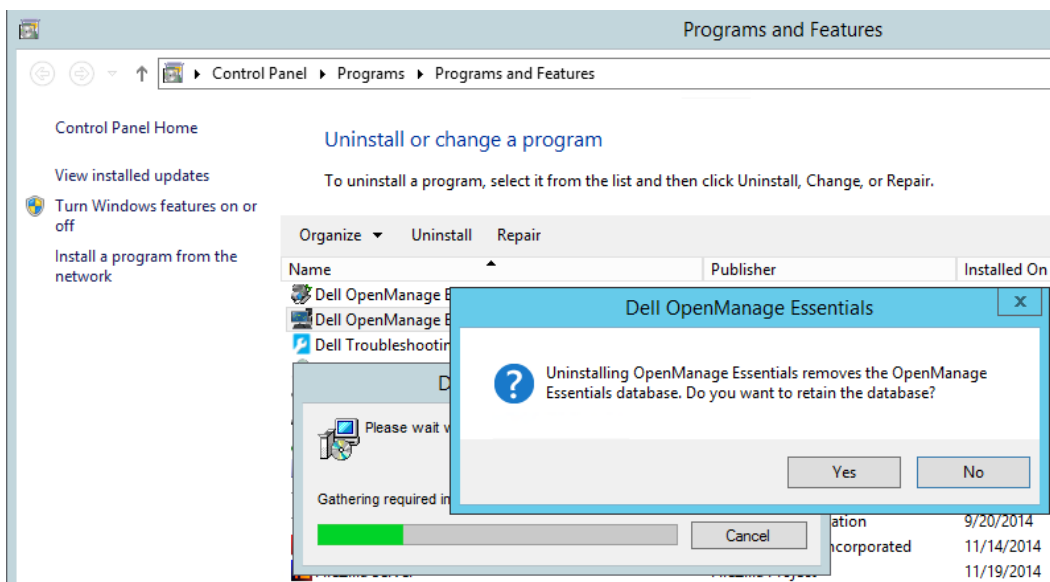


Note: As *Dell SupportAssist* and *Dell MIBImport Utility* functionality are dependent upon OME, uninstallation of OME checks their presence. You must first uninstall *Dell SupportAssist* and *Dell MIB Import Utility* to uninstall OME.

Note: Uninstalling OpenManage Essentials does not remove the Troubleshooting Tool. It needs to be uninstalled separately.

Note: OME uninstallation lets you retain the database (Figure 16). You can use the database again with the same version or after upgrading to a future version of OpenManage Essentials.

Figure 16. Retain Database Option while Uninstalling



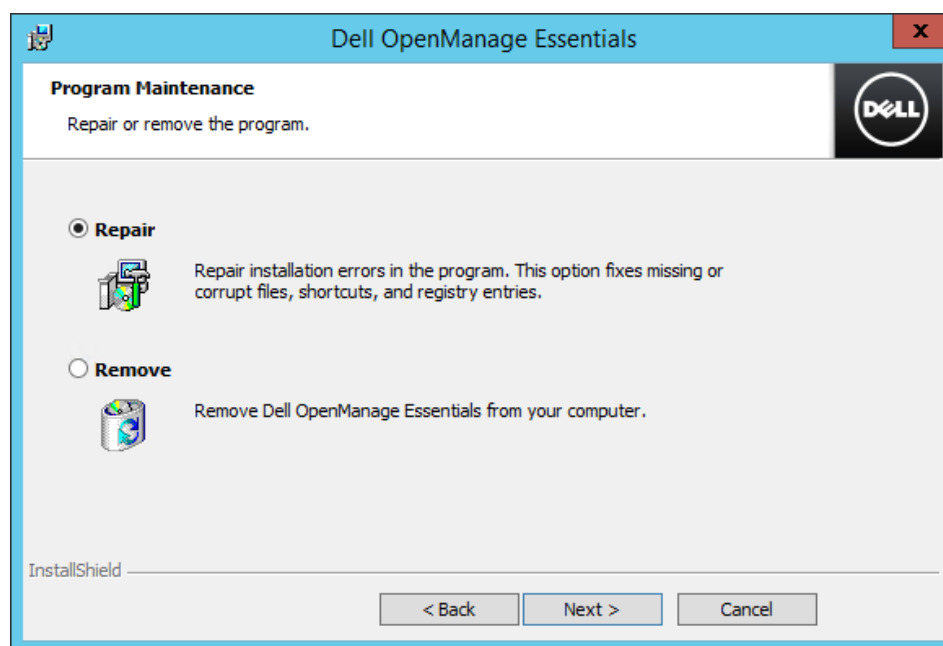
Repair (Program Maintenance)

Installing Dell OpenManage Essentials

You can repair OpenManage Essentials by going to the Program maintenance window. Do this if the installation drive or folder is corrupted or if you have accidentally deleted the OME installed files. This lets you recreate the database (fresh database) or use an existing one (if you already have the database files).

Note: If the OME database is corrupted, it is recommended that you recreate the database during repair or install a completely new OpenManage Essentials instance.

Figure 17. Program maintenance.



Learn more

To know about OpenManage Essentials, refer www.delltechcenter.com/ome

Conclusion

OpenManage Essentials is an easy-to-install, one-to-many console that simplifies systems management with features like single-click prerequisites installation. Additionally, it allows Remote Database installation.